AI AND PROMPT ENGINEERING

UNIT 4 · AI APPLICATIONS ACROSS ENGINEERING DOMAINS

AKTU B.Tech First Year | AAI101 / AAI201 | 2026–27 Scheme

COMPLETE STUDY NOTES

Eight syllabus areas · simple explanations · working flowcharts · comparison tables

SOLVED QUESTION BANK

Section A (2 marks) · Section B (7 marks) · Section C (short notes and differences) · Section D (case studies)

Designed for clear learning and exam-focused revision

Colour palette: blush pink · muted rose · warm cream

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Study tip: For a 2-mark answer, lead with the definition and add two precise points. For a 7-mark answer, use a definition, working/flowchart, applications, advantages, limitations and a short conclusion.

How to write strong exam answers

- Expand an abbreviation the first time it appears, then use the abbreviation consistently.
- Use a small labelled flowchart when a question asks how a system works.
- Give a concrete example, but do not invent statistics or claim that AI is always correct.
- Mention human review, data quality and privacy when discussing high-impact AI systems.

1. AI in Civil Engineering

Definition: AI in civil engineering is the use of machine-learning, computer-vision and data-analysis methods to plan, monitor, maintain and operate buildings, bridges, roads and urban infrastructure.

1.1 Structural health monitoring (SHM)

Concept: Structural health monitoring checks the condition and safety of structures over time. Sensors may measure vibration, strain, tilt, temperature or crack growth. AI analyses these readings to identify unusual patterns that may indicate damage.

Sensors on bridge/building → Collect vibration, strain and crack data → AI detects abnormal patterns → Engineer inspects the structure → Repair or monitor

Applications: bridge monitoring, earthquake-damage assessment, tunnel inspection, dam safety and monitoring large buildings.

Advantages: earlier warning; continuous monitoring; helps prioritise inspections; can reduce unexpected repair costs.

Limitations: sensor failures, noisy data, high installation cost and false alarms. AI predictions should support—not replace—qualified engineering inspection.

1.2 Smart cities and urban planning

A **smart city** uses connected sensors, digital systems and data analysis to improve city services. AI can combine information about energy, water, transport, waste and public infrastructure to support planning.

- Forecast demand for electricity and water.
- Optimise waste collection routes and public transport schedules.
- Detect water leaks or unusual energy consumption.
- Support land-use planning using satellite images and geographic information systems (GIS).

1.3 Traffic optimisation

AI uses traffic-camera images, road sensors, GPS and historical traffic data to estimate congestion and adjust traffic signals or route recommendations.

Camera/GPS/road sensors → Traffic data → AI estimates congestion → Signals/routes adjusted → Monitor traffic outcome

Example: an adaptive signal system can extend a green phase when one approach has a longer queue, subject to safety rules and pedestrian requirements.

Exam keywords

Structural health monitoring · computer vision · sensor fusion · smart city · adaptive traffic signals · predictive planning

2. AI in Mechanical Engineering

Definition: AI in mechanical engineering applies data-driven models, optimisation and intelligent control to machines, product design, manufacturing and robots.

2.1 Predictive maintenance

Predictive maintenance estimates when equipment may fail by learning patterns in sensor data. It differs from simply repairing equipment after breakdown.

Sensors (vibration, heat, sound) → Data collection → AI model identifies patterns → Failure-risk alert → Planned maintenance → Feedback updates model

Working: (1) collect normal and failure-related data; (2) clean and label data; (3) train a model; (4) estimate fault type or risk; (5) alert maintenance staff; (6) verify the machine and feed outcomes back into the system.

Applications: pumps, motors, turbines, bearings, CNC machines, factory conveyors and aircraft components.

Advantages: reduced unplanned downtime, better spare-parts planning, improved equipment life and safer maintenance scheduling.

Limitations: needs reliable historical data, sensors and skilled staff; a model can miss new fault types.

2.2 Design optimisation

AI-assisted design explores many design alternatives and predicts their performance against constraints such as weight, strength, cost, energy use and manufacturability. Engineers still define the objectives and validate the selected design.

- Generative design proposes candidate shapes or layouts.
- Surrogate models estimate simulation results more quickly.
- Optimisation algorithms search for designs meeting multiple constraints.
- Examples: lightweight vehicle components, heat exchangers and efficient turbine blades.

2.3 Robotics and intelligent automation

AI enables robots to perceive their environment, recognise objects, plan actions and adapt to changing conditions. Industrial robots may use computer vision for pick-and-place, inspection, welding and assembly.

Camera/sensors → Object or scene recognition → Motion planning → Robot action → Safety check and feedback

Limitations: setup cost, safety risks around people, difficulty with unexpected objects and the need for maintenance and retraining.

Exam keywords

Predictive maintenance · condition monitoring · generative design · optimisation constraints · computer vision · industrial robotics

3. AI in Electrical and Electronics Engineering

Definition: AI in electrical and electronics engineering uses data analysis and learning algorithms to manage electrical networks, detect faults and assist electronic-system design.

3.1 Smart grids

A **smart grid** is an electricity network that uses digital communication, sensors and automated control to monitor and manage power generation, distribution and consumption.

Smart meters + grid sensors → Live demand and supply data → AI forecasts load/renewables → Control recommendations → Operator or automated controller acts

- Forecast peak electricity demand.
- Balance variable solar and wind generation with demand and storage.
- Detect abnormal losses and outages.
- Support demand response, where consumers shift usage away from peak periods.

Benefits: improved reliability, more efficient energy use, faster outage detection and better integration of renewable energy.

Limitations: cybersecurity risks, communication failures, biased or incomplete data and the need for reliable fail-safe controls.

3.2 Fault detection and diagnosis

AI compares measurements such as voltage, current, temperature and harmonics with learned or expected patterns to identify likely faults. It can help detect overheating, insulation problems, short circuits or abnormal power quality.

Voltage/current/temperature readings → Feature extraction → AI classifies anomaly/fault → Alert and location estimate → Technician verifies and repairs

AI output is a diagnostic aid. Protective relays and engineered safety mechanisms must remain dependable even if the AI system fails.

3.3 VLSI design automation

Very Large-Scale Integration (VLSI) places many electronic components on one chip. AI can assist chip design by predicting timing, power and area; exploring placement and routing options; and identifying design-rule problems.

- Speeds up exploration of design alternatives.
- Can improve power-performance-area trade-offs.
- Helps prioritise verification and detect patterns in design failures.
- Final designs still require formal verification, testing and engineering sign-off.

Exam keywords

Smart grid · load forecasting · demand response · fault diagnosis · VLSI · power-performance-area (PPA)

4. AI in Computer Science and Engineering (CSE)

Definition: AI in CSE is used to create software, detect digital threats, understand human language and automate selected computing tasks.

4.1 AI-based code generation

Tools such as **GitHub Copilot** use context from prompts, comments and nearby code to suggest code, tests or explanations. Developers review, test and modify the suggestions.

Developer prompt + code context → AI generates suggestion → Developer reviews → Tests/security checks → Accept, edit or reject

Advantages: can reduce repetitive typing, explain unfamiliar code and suggest tests. **Limitations:** may produce incorrect, insecure, outdated or licence-sensitive code; generated code is not proof of correctness.

4.2 Cybersecurity and threat detection

AI analyses network traffic, login patterns, files and system logs to identify suspicious behaviour. It can detect anomalies or classify known malware patterns and help prioritise alerts.

- Flag unusual login locations or access times.
- Identify suspicious network traffic and phishing indicators.
- Group alerts to help analysts investigate incidents.
- Limitations include false positives, adversarial evasion and privacy concerns.

4.3 Natural language interfaces

A natural language interface lets a person interact with a computer using ordinary language, either by text or speech. Chatbots and voice assistants use natural language processing (NLP), language models and sometimes speech recognition.

User text/speech → Language processing → Intent/context interpretation → Response or tool action → User feedback

Such systems are useful for customer support, search, accessibility and question-answering. They may misunderstand ambiguity, invent information or act on misleading instructions, so important actions need confirmation.

Exam keywords

Code generation · human-in-the-loop · cybersecurity · anomaly detection · NLP · conversational AI

5. AI in Biotechnology and Healthcare

Definition: AI in biotechnology and healthcare analyses medical, biological and patient-generated data to support research, diagnosis, monitoring and treatment planning.

5.1 Drug discovery and development

AI can rank candidate molecules, predict selected molecular properties, analyse biological targets and help identify promising compounds for laboratory testing.

Biological target + research data → AI screens/predicts candidates → Scientists select compounds → Laboratory tests → Clinical trials and regulatory review

AI may narrow a search space, but it does not by itself prove that a drug is safe or effective. Laboratory experiments, clinical trials and regulatory review remain essential.

5.2 Medical imaging and disease detection

Computer-vision models analyse X-rays, CT scans, MRI scans, ultrasound or retinal images to highlight patterns associated with possible disease. They can help clinicians prioritise cases or measure changes over time.

Medical image → Quality check → AI highlights possible finding → Clinician reviews with patient history → Diagnosis/next step

Benefits: assistance with image review, consistency and prioritisation. **Limitations:** performance may vary across hospitals and populations; image quality and clinical context matter.

5.3 Genomics and genetic-data analysis

AI can help identify patterns in DNA sequences, classify genetic variants, compare gene-expression data and explore relationships between biological features and disease. Results require expert interpretation and validation.

5.4 Wearables and health monitoring

Smartwatches and other wearable devices collect measures such as heart rate, activity and sleep-related signals. AI may detect trends or irregular patterns and prompt a user to seek advice.

Wearable alerts are not definitive diagnoses. Devices can be inaccurate, and health data requires strong privacy and security protections.

Exam keywords

Drug candidate screening · medical imaging · computer vision · genomics · wearable sensors · clinical validation

6. AI in Agricultural Engineering

Definition: AI in agricultural engineering uses data from fields, satellites, cameras, drones and soil sensors to improve crop management and resource use.

6.1 Precision farming

Precision farming manages a field according to local conditions instead of treating every part identically. AI combines soil, weather, crop and location data to recommend where and when to irrigate, fertilise or inspect.

Soil/weather/crop data → AI maps field variability → Recommendation by zone → Targeted irrigation/fertiliser/action → Measure results

6.2 Crop monitoring and disease detection

AI models analyse leaf or crop images to identify possible disease, pest damage, nutrient stress or water stress. A farmer or agronomist confirms the result before treatment.

6.3 Drone-based spraying

Drones with cameras or mapping tools can locate affected areas. Where legally permitted and safe, targeted spraying may reduce unnecessary chemical application compared with blanket treatment.

6.4 Soil analysis and intelligent irrigation

Soil sensors may measure moisture, pH, temperature or electrical conductivity. AI combines readings with weather forecasts and crop needs to recommend irrigation timing and amount.

Benefits: more targeted inputs, water-use efficiency, earlier crop-stress detection and improved field records.
Limitations: cost, connectivity, battery life, sensor calibration, model errors and access to local training data.

Exam keywords

Precision agriculture · computer vision · soil moisture · targeted spraying · remote sensing · intelligent irrigation

7. AI in Chemical and Textile Engineering

Definition: AI in chemical and textile engineering uses process data, optimisation and image analysis to improve production efficiency, material development and product quality.

7.1 Process optimisation

In chemical plants, AI can model relationships among temperature, pressure, flow rate and product quality. It may recommend operating settings that balance yield, energy use, cost and safety.

Process sensors + lab results → AI estimates process state/quality → Suggested operating adjustment → Operator/control system checks safety → Monitor outcome

AI must operate within engineering limits; a recommendation should never override safety interlocks or process-safety procedures.

7.2 Quality control and defect detection

Computer vision inspects products for visible defects. In textiles it can flag holes, stains, broken threads, uneven patterns or colour variation; in chemical manufacturing, AI can detect unusual sensor or laboratory patterns that indicate off-specification output.

7.3 AI-based material design

AI models can predict selected properties of candidate polymers, composites, coatings or fibres, helping researchers choose promising materials for experiments. Predictions require laboratory validation.

Area	AI use	Illustrative example
Chemical engineering	Process monitoring and optimisation	Predicting a quality deviation from temperature and flow data
Textile engineering	Visual defect detection	Camera flags a weave defect on moving fabric
Materials	Property prediction	Ranking polymer formulations for further testing

Benefits: consistent quality, less waste, faster experimentation and better process visibility. **Limitations:** sensor drift, changing raw materials, poor labels and the cost of integration.

Exam keywords

Process optimisation · quality control · defect detection · materials informatics · process safety

8. Ethical Concerns Related to AI

Definition: AI ethics concerns the principles and practices used to make AI systems fair, safe, transparent, privacy-preserving and accountable to people.

8.1 Algorithmic bias and fairness

Algorithmic bias occurs when an AI system produces systematically unfair outcomes for particular people or groups. Bias can enter through unrepresentative data, historical discrimination, poor labels or proxy variables.

Example: a loan model may reject more applicants from a particular area because historical data reflects unequal access to credit. Location can act as a proxy for other sensitive characteristics.

Risk reduction: examine data quality and group-level outcomes; test for disparate errors; improve representation; document limitations; use human review and appeal routes.

8.2 Data privacy and security

Privacy concerns appropriate collection and use of personal data. **Security** concerns protection against unauthorised access, alteration, loss or disclosure. A system may be secure but still collect more data than necessary.

- Collect only data needed for a defined purpose.
- Obtain valid consent where required and explain data use.
- Use access controls, encryption, retention limits and secure deletion.
- Anonymise or minimise personal information where appropriate.
- Test for data leakage and restrict access to sensitive datasets.

8.3 Job displacement and employment

Automation may reduce demand for some repetitive tasks while changing other jobs and creating new roles. The impact differs by sector, task and adoption speed.

- Provide reskilling and upskilling opportunities.
- Redesign work so people handle judgement, communication and oversight.
- Plan transitions fairly and involve workers in deployment decisions.
- Assess productivity benefits alongside effects on workers and communities.

8.4 Responsible AI principles

Principle	Meaning in practice
Fairness	Check for unjustified differences in outcomes across groups.
Transparency	Explain purpose, important factors, limitations and uncertainty in understandable terms.
Accountability	Assign people or organisations responsibility for outcomes and remedies.
Privacy and security	Minimise data collection and protect information throughout its life cycle.
Safety and reliability	Test performance, monitor failures and provide safe fallback behaviour.
Human oversight	Keep meaningful review and appeal for high-impact decisions.
Inclusiveness	Consider accessibility and the needs of affected users.

8.5 Explainable AI and human oversight

Explainable AI (XAI) refers to methods that help people understand why a model produced a particular output. An explanation should be appropriate to the audience and should not be treated as proof that the model is correct.

Responsible deployment includes pre-release testing, documentation, monitoring after release, incident reporting, user feedback and a way to challenge important decisions.

Exam keywords

Algorithmic bias · fairness · data minimisation · privacy · cybersecurity · transparency · accountability · explainability · human oversight

9. Quick Revision Sheet

Topic	One-line summary	Keywords to remember
Civil	AI monitors structures and helps plan cities and traffic.	SHM, sensors, smart city, adaptive signals
Mechanical	AI predicts machine faults, optimises designs and supports robots.	Predictive maintenance, optimisation, robotics
Electrical/electronics	AI forecasts grid demand, detects faults and assists chip design.	Smart grid, load forecast, diagnosis, VLSI
CSE	AI generates code suggestions, detects threats and enables language interfaces.	Copilot, anomaly detection, NLP
Biotechnology/healthcare	AI supports drug research, imaging, genomics and wearable monitoring.	Screening, computer vision, genomics, validation
Agriculture	AI guides field-specific crop care, disease checks and irrigation.	Precision farming, drones, soil sensors
Chemical/textile	AI optimises processes, detects defects and predicts material properties.	Process control, quality, material design
Ethics	AI must be fair, private, secure, accountable and supervised.	Bias, privacy, transparency, human oversight

Essential definitions

- **AI:** techniques that enable computer systems to perform tasks associated with human intelligence.
- **Machine learning (ML):** methods that learn patterns from data to make predictions or decisions.
- **Computer vision:** methods that interpret information from images or video.
- **Predictive maintenance:** using condition data and models to estimate failure risk and plan maintenance.
- **Smart grid:** an electricity network using digital monitoring, communication and control.
- **Precision agriculture:** managing crops and inputs according to local field conditions.
- **Digital twin:** a digital representation of a physical asset or process that is updated with relevant data; it may support monitoring and simulation.
- **Algorithmic bias:** systematic unfairness in an algorithm's outcomes.
- **Responsible AI:** developing and using AI with fairness, safety, privacy, transparency, accountability and human oversight.

Useful general AI application pattern

Real-world problem → Collect relevant data → Prepare/check data → Train or select AI model → Validate results → Human/automated decision → Monitor and improve

Remember: data quality affects output quality. AI is a decision-support tool in many engineering settings; critical systems require verification, safety controls and expert oversight.

Last-minute checklist

- Can I define all eight application areas in 1–2 sentences?
- Can I draw one flowchart for predictive maintenance, smart grids, medical imaging and precision farming?
- Can I list at least three applications and two limitations for each major domain?
- Can I distinguish preventive from predictive maintenance, privacy from security, and manual from AI-based inspection?
- Can I explain bias, responsible AI and human oversight with an example?

10. Additional Exam Practice

These are practice prompts based on the supplied syllabus, not an official or guaranteed AKTU question prediction.

- Define structural health monitoring and explain its AI-enabled workflow.
- Explain how predictive maintenance differs from preventive maintenance.
- Describe AI's role in a smart grid with a labelled flowchart.
- Discuss AI in medical imaging and explain why clinical validation is necessary.
- Explain precision agriculture with examples of crop monitoring and intelligent irrigation.
- Discuss algorithmic bias, privacy risks and three ways to support responsible AI.
- Write a short note on AI-based quality inspection in textile manufacturing.
- Explain how AI can assist VLSI design automation.

11. Solved Question Bank — Section A

Short-answer questions · 2 marks each. Original numbering is preserved from the supplied question bank.

1. (Imp.) Give two applications of AI in civil engineering.

1. **Structural health monitoring:** AI analyses vibration or strain data to flag possible damage in bridges and buildings.
2. **Traffic optimisation:** AI analyses traffic flow and helps adjust signal timing or suggest routes.

2. What is structural health monitoring?

Structural health monitoring (SHM) is the continuous or periodic assessment of a structure's condition using sensors and inspection data. AI can identify unusual vibration, strain or crack patterns that may require an engineer's inspection.

3. (Imp.) What is predictive maintenance?

Predictive maintenance uses equipment-condition data and AI models to estimate the risk or timing of failure. Maintenance is planned before a likely breakdown, reducing unplanned downtime.

4. How is AI used in manufacturing quality inspection?

AI, especially computer vision, analyses images or sensor readings to detect defects such as cracks, stains, incorrect dimensions or textile flaws. It can flag suspicious items for rejection or human review.

5. Give two applications of AI in mechanical engineering.

1. Predictive maintenance of motors, pumps and turbines.
2. Design optimisation to reduce weight, cost or energy use while meeting strength and safety constraints.

6. (Imp.) How does AI help in a smart power grid?

AI forecasts electricity demand, helps balance renewable supply and demand, and detects abnormal grid conditions. This supports reliable and efficient power distribution.

7. Give two applications of AI in electronics and communication.

1. VLSI/chip design optimisation, such as exploring placement and routing choices.
2. Fault or anomaly detection in electronic equipment and communication networks.

8. Give two applications of AI in computer science and engineering.

1. AI-based code suggestions and test generation, such as GitHub Copilot.
2. Cybersecurity monitoring to identify suspicious network activity or possible malware.

9. How is AI used in medical imaging?

AI models analyse X-rays, CT, MRI or retinal images to highlight patterns that may indicate disease. Clinicians review these outputs with patient history before making a diagnosis.

10. How does AI speed up drug discovery?

AI screens large sets of candidate molecules and predicts selected properties, helping researchers prioritise compounds for laboratory testing. It does not replace experiments or clinical trials.

11. (Imp.) What is precision agriculture?

Precision agriculture uses data from soil sensors, satellites, drones and crop images to manage a field according to local needs, improving the targeting of water, fertiliser and other inputs.

12. How can AI detect crop disease?

A computer-vision model compares a crop or leaf image with patterns learned from labelled examples and flags a possible disease. A farmer or agronomist should confirm the result before treatment.

13. Give one application of AI each in chemical and textile engineering.

Chemical: predicting process-quality deviations from temperature, pressure and flow data.
Textile: using camera images to detect fabric holes, stains or weaving defects.

14. What is a digital twin?

A digital twin is a digital representation of a physical asset, system or process that is connected to relevant data. It can support monitoring, simulation and maintenance planning.

15. (Imp.) What is algorithmic bias? Give an example.

Algorithmic bias is systematic unfairness in an AI system's outcomes. For example, a loan model trained on historically biased decisions may unfairly reject qualified applicants from certain areas.

16. Why is data privacy a concern in AI?

AI may use sensitive personal, health, location or financial data. Excessive collection, unclear consent or unauthorised sharing can harm people, so data use must be limited and protected.

17. What is meant by job displacement due to AI?

Job displacement occurs when automation reduces the need for people to perform certain tasks or roles. Reskilling and redesigning work can help workers adapt to changing job requirements.

18. (Imp.) What is responsible AI?

Responsible AI is the design and use of AI systems with fairness, safety, privacy, transparency, accountability and appropriate human oversight.

19. What is explainable AI?

Explainable AI (XAI) refers to methods that help people understand the factors or reasoning behind a model's output. Explanations aid review but do not guarantee correctness.

12. Solved Question Bank — Section B

Long-answer questions · 7 marks each. Answers are structured for exam writing.

1. (Imp.) Explain the common pattern of AI applications in engineering (sensor → data → model → decision) with examples.

Introduction: Across engineering fields, AI turns observations from real systems into predictions or recommendations. The common pattern is data collection, processing, model analysis, decision and feedback.
Working:
1. **Sense/collect:** obtain measurements from sensors, cameras, logs or records.
2. **Prepare data:** remove errors, align timestamps and select useful features.
3. **Model:** train or apply an AI model to classify, forecast or detect anomalies.
4. **Decision:** send an alert or recommendation to an engineer/operator, or to a validated controller.
5. **Verify and learn:** check the outcome and monitor performance over time.
Examples: vibration sensors predict pump failure; smart-grid data forecasts peak demand; leaf images flag crop disease; cameras detect textile defects.
Benefits: earlier detection, faster analysis, more consistent decisions and better use of resources.
Limitations: poor data, model error, cyber risks and changing conditions can affect performance. Safety-critical decisions require validation and human oversight.
Conclusion: AI is most useful when integrated with reliable data, engineering knowledge and a feedback process.

2. Discuss applications of AI in civil engineering.

Introduction: AI supports the planning, construction, monitoring and maintenance of infrastructure.
Applications:
1. **Structural health monitoring:** analyse vibration, strain and crack data to flag possible damage in bridges, dams and buildings.
2. **Smart cities:** forecast water and energy demand, plan waste collection and support infrastructure management.
3. **Traffic optimisation:** use camera, GPS and sensor data to estimate congestion and adjust traffic signals.
4. **Construction monitoring:** analyse site images to track progress, identify hazards or compare work against plans.
5. **Urban planning:** use satellite images and geographic data to support land-use and transport decisions.
Flowchart: Infrastructure sensors/images → Data analysis → AI detects patterns or predicts demand → Engineer/city operator reviews → Action and monitoring.
Advantages: early warnings, improved planning, reduced delays and better resource allocation.
Limitations: sensor and data quality, implementation cost, privacy concerns and risk of false alerts.
Conclusion: AI improves civil-engineering decisions when combined with site inspection, standards and professional judgement.

3. (Imp.) Discuss applications of AI in mechanical engineering, especially predictive maintenance.

Introduction: AI helps mechanical engineers monitor machines, improve designs and automate manufacturing.
Predictive maintenance: Sensors collect vibration, temperature, sound or current data. An AI model learns patterns linked with normal operation and faults. When risk increases, it alerts maintenance staff so the machine can be checked before failure.
Flowchart: Sensors → Data collection and cleaning → AI fault-risk model → Alert → Inspection and planned maintenance → Feedback.
Other applications:
1. **Design optimisation:** compare shapes and materials against weight, cost, strength and energy constraints.
2. **Robotics:** use vision and control for assembly, welding, handling and inspection.
3. **Manufacturing:** identify defects and improve production scheduling.
4. **Energy efficiency:** analyse machine operation and recommend adjustments.
Advantages: less unplanned downtime, longer equipment life, improved quality and more efficient designs.
Limitations: sensors and data are costly; new fault patterns may be missed; staff must verify alerts.
Conclusion: predictive maintenance is valuable because it changes maintenance from reacting after failure to planning based on condition and risk.

4. Discuss applications of AI in electrical and electronics engineering.

Introduction: AI supports electrical networks, equipment diagnosis and electronic-chip design.
1. **Smart grids:** forecast demand, balance supply and demand, integrate renewable energy and help detect outages.
2. **Fault detection:** analyse voltage, current, temperature and waveform patterns to identify likely abnormal conditions in motors, transformers or circuits.
3. **VLSI design automation:** help explore chip placement, routing, timing, power and area trade-offs.
4. **Energy management:** predict building loads and recommend efficient operation of electrical equipment.
5. **Communication systems:** detect unusual traffic patterns and support network optimisation.
Flowchart: Smart meters/sensors → Data collection → AI forecasting or diagnosis → Operator/control response → Monitoring.
Advantages: faster fault awareness, improved reliability, better energy use and quicker design exploration.
Limitations: cyber risks, incomplete data, false alarms and the need for verification. Protective relays and safety interlocks remain essential.
Conclusion: AI can make electrical and electronic systems more adaptive, provided engineering safeguards remain in place.

5. Discuss applications of AI in computer science and engineering.

Introduction: AI is used to assist software development, cybersecurity and human-computer interaction.
1. **Code generation:** tools such as GitHub Copilot suggest code, tests and explanations from prompts and surrounding context. Developers must review and test outputs.
2. **Cybersecurity:** models analyse logs, network traffic and login behaviour to flag suspicious activity, possible malware or phishing.
3. **Natural language interfaces:** chatbots and voice assistants interpret user requests and provide answers or invoke tools.
4. **Software testing:** AI can suggest test cases, classify bugs and summarise logs.
5. **Data analysis:** AI can classify, forecast and detect anomalies in large datasets.
Advantages: automates repetitive work, accelerates analysis and improves accessibility.
Limitations: generated code can be wrong or insecure; systems may produce false alarms or fabricated answers; privacy and intellectual-property concerns require care.
Conclusion: AI is an assistant, not a substitute for software engineering fundamentals, testing and security review.

6. (Imp.) Discuss the role of AI in biotechnology and healthcare.

Introduction: AI analyses biological, medical and patient data to support research, diagnosis and monitoring.
1. **Drug discovery:** screen candidate molecules and predict selected properties to prioritise laboratory experiments.
2. **Medical imaging:** analyse X-rays, CT, MRI or retinal images and highlight possible abnormalities for clinician review.
3. **Genomics:** find patterns in DNA sequences, gene expression and genetic variants.
4. **Wearables:** analyse heart-rate or activity trends and flag possible irregular patterns.
5. **Hospital operations:** forecast resource needs and help organise workflows, subject to privacy and fairness controls.
Flowchart: Validated data → AI analysis → Candidate finding or alert → Expert review → Testing/clinical decision → Monitoring.
Benefits: faster screening, support for image review, personalised research and continuous monitoring.
Limitations: biased datasets, privacy risks, errors across populations and lack of clinical context. Drug candidates require laboratory tests and clinical trials; diagnostic outputs require clinician judgement.
Conclusion: AI can support healthcare and biotechnology, but validation, patient safety and data protection are essential.

7. Discuss applications of AI in agricultural engineering.

Introduction: AI helps farmers make decisions using crop, soil, weather and field data.
1. **Precision farming:** divide fields into zones and apply water or fertiliser according to local needs.
2. **Crop monitoring:** analyse satellite, drone or phone images to track growth and stress.
3. **Disease detection:** computer vision flags possible disease patterns on leaves or fruit.
4. **Drone-based spraying:** map affected areas for targeted application where permitted and safe.
5. **Intelligent irrigation:** combine soil-moisture readings, crop needs and weather forecasts to recommend watering.
Flowchart: Soil/weather/camera data → AI analyses crop and field conditions → Recommendation/map → Farmer checks → Targeted action → Results monitored.
Advantages: efficient water use, reduced unnecessary inputs, earlier detection and improved records.
Limitations: cost, connectivity, sensor calibration, local-data shortages and possible model error.
Conclusion: AI is most effective when recommendations are locally validated and combined with farmer expertise.

8. Discuss applications of AI in chemical and textile engineering.

Introduction: AI can improve process performance, product quality and materials research.
Chemical engineering:
1. Analyse temperature, pressure and flow to predict quality deviations.
2. Recommend operating settings that balance yield, energy, cost and safety.
3. Detect abnormal process behaviour for operator investigation.
Textile engineering:
1. Use computer vision to detect holes, stains, broken threads and uneven patterns.
2. Monitor colour consistency and fabric dimensions.
3. Support production planning and reduce material waste.
Material design: predict selected properties of polymers, fibres, coatings or composites to prioritise experiments.
Flowchart: Process sensors/images → AI analysis → Quality or optimisation recommendation → Human/safety checks → Process adjustment and monitoring.
Advantages: improved consistency, reduced waste, faster inspection and more focused research.
Limitations: changing raw materials, sensor drift, poor labels, integration cost and the need for laboratory validation.
Conclusion: AI can assist production and quality decisions, but process safety and verification must remain central.

9. (Imp.) Discuss the ethical concerns of AI: bias, data privacy and job displacement. How can they be reduced?

1. Algorithmic bias: If training data reflects historical discrimination or excludes some groups, model outputs may be unfair. **Example:** a loan model rejects qualified applicants from a particular area because location acts as a proxy feature. **Reduction:** audit datasets and group-level outcomes, improve representation, test error rates, document limitations and provide human review and appeals.
2. Data privacy and security: AI may process sensitive personal, health or financial data. Excessive collection, unclear consent or breaches can cause harm. **Reduction:** collect only necessary data, explain purposes, use access controls and encryption, limit retention and test for data leakage.
3. Job displacement: automation can reduce demand for some tasks and change job roles. **Reduction:** offer reskilling, redesign roles around human judgement, involve workers and plan transitions fairly.
Other principles: transparency, accountability, safety, explainability and human oversight.
Conclusion: ethical AI requires continuous testing and governance, not just accurate predictions.

10. What are the principles of responsible AI?

Responsible AI means designing, deploying and monitoring AI so it benefits people and avoids foreseeable harm. Key principles are:
1. **Fairness:** check that outputs do not create unjustified discrimination.
2. **Transparency:** explain the system's purpose, limitations and important factors in understandable terms.
3. **Accountability:** assign clear responsibility for decisions, monitoring and remedies.
4. **Privacy and security:** minimise personal data and protect it from unauthorised access.
5. **Safety and reliability:** test before deployment, monitor failures and maintain safe fallback options.
6. **Human oversight:** ensure people can review, challenge or override high-impact outputs.
7. **Inclusiveness and accessibility:** consider diverse users and affected communities.

Implementation: document datasets and models, run bias and security tests, monitor real-world performance, report incidents and update the system when conditions change.
Conclusion: responsible AI combines technical quality with human rights, governance and ongoing accountability.

11. “AI will not replace engineers, but engineers who use AI will replace those who don't.” Discuss.

The statement highlights that AI is likely to automate parts of engineering work, while engineers remain responsible for defining problems, checking results and making safety-critical judgements.
How AI supports engineers:
1. Analyses large datasets faster than manual review.
2. Helps predict equipment failure and plan maintenance.
3. Generates design options and supports optimisation.
4. Assists with coding, documentation and testing.
5. Detects patterns in images, sensor data and production records.

Why engineers remain important: they understand requirements, constraints, standards, context, ethics and consequences. AI can be inaccurate, biased or unsuitable for unfamiliar conditions.
Challenges: engineers need AI literacy, data skills, critical thinking, cybersecurity awareness and continuous learning. Over-reliance without verification can create serious errors.
Conclusion: AI is best treated as a productivity tool. Engineers who combine domain knowledge with responsible use of AI may work more effectively, but the outcome depends on the task, training and organisation—not on AI use alone.

13. Solved Question Bank — Section C

Short notes and differences · 3–4 marks each.

1. (Imp.) Short note: Predictive maintenance

Predictive maintenance uses sensor data and AI models to estimate the likelihood of equipment failure. Sensors may measure vibration, temperature, sound or current. The model identifies patterns associated with faults and alerts staff so inspection or repair can be planned. It can reduce unexpected downtime and improve spare-parts planning. Its effectiveness depends on reliable data and engineering verification.

2. Short note: Smart grid

A smart grid is an electricity network that uses sensors, smart meters, digital communication and automated control. AI can forecast demand, help integrate renewable energy, detect anomalies and support outage management. It improves monitoring and energy efficiency, but must be protected against cyberattacks and communication failures.

3. Short note: AI in medical imaging

AI models analyse X-rays, CT, MRI, ultrasound or retinal images to highlight patterns associated with possible disease. They can help prioritise scans and support clinicians, but image quality, patient history and population differences affect performance. The clinician remains responsible for interpreting the result.

4. Short note: Precision agriculture

Precision agriculture uses field-specific information from soil sensors, weather services, satellites, drones and crop images. AI helps identify crop stress, map disease, and recommend targeted irrigation or fertiliser. It can reduce unnecessary inputs, but requires suitable data, affordable technology and local validation.

5. (Imp.) Short note: Algorithmic bias

Algorithmic bias is systematic unfairness in an AI system's outputs. It may result from unrepresentative data, historical discrimination, poor labels or proxy variables. For example, a loan model may unfairly disadvantage applicants from certain areas. Audits, balanced data, group-level testing, transparency and human appeals can reduce the risk.

Differentiate between

Basis	Preventive maintenance	Predictive maintenance
Meaning	Maintenance is performed on a fixed schedule or after a set amount of use.	Maintenance is planned using condition data and estimated failure risk.
Trigger	Time, calendar or usage interval.	Sensor readings, model output or observed condition.
Data need	May require little live sensor data.	Usually depends on condition data and a validated model.
Strength	Simple to plan and useful where schedules are known.	Can target maintenance when it is more likely to be needed.
Limitation	May service healthy equipment too early or miss failures between intervals.	Needs reliable data and can make incorrect predictions.

1. (Imp.) Preventive and predictive maintenance — conclusion: preventive maintenance is schedule-based, while predictive maintenance is condition- and risk-based.

Basis	Manual quality inspection	AI-based quality inspection
Method	Human visually or physically checks products.	Model analyses images or sensor data and flags likely defects.
Speed	Can be slow for high-volume production.	Can process many items quickly when integrated well.

Basis	Manual quality inspection	AI-based quality inspection
Consistency	May vary with fatigue and individual judgement.	Can apply consistent criteria, but model errors remain possible.
Best use	Complex contextual judgement and unusual defects.	Repetitive, measurable patterns and high-throughput screening.
Limitations	Labour-intensive; fatigue can affect results.	Needs good training data, calibration and human review.

2. Manual and AI-based quality inspection — conclusion: AI can improve speed and consistency, while humans remain valuable for ambiguous cases and oversight.

Basis	Data privacy	Data security
Focus	Appropriate collection, use and sharing of data.	Protection of data against unauthorised access, change, loss or disclosure.
Main question	Should this data be collected or used, and for what purpose?	Is the data adequately protected from threats?
Controls	Consent where required, data minimisation, purpose limitation, retention limits.	Encryption, access control, authentication, backups and monitoring.
Relationship	Privacy can be violated even when systems are secure.	Security helps protect privacy but does not guarantee appropriate use.

3. Data privacy and data security — conclusion: privacy governs appropriate data use; security protects data against threats.

14. Solved Question Bank — Section D

Case-based questions. The original case labels D1–D3 are preserved.

D1. (Imp.) A factory's pumps fail without warning, stopping production. Suggest an AI solution.

Solution: Predictive maintenance.
1. Fit vibration, temperature and, where useful, current sensors to the pumps.
2. Collect data during normal operation and from known faults; label and clean the data.
3. Train and validate a model to recognise patterns that tend to occur before failure.
4. Monitor live readings and estimate abnormality or failure risk.
5. Alert maintenance staff early so they can inspect the pump and plan repairs.
6. Record the confirmed fault and maintenance outcome to improve future monitoring.

Flowchart: Sensors → Data → AI fault-risk model → Early alert → Inspection and planned repair.
Benefits: less unplanned downtime, lower emergency-repair burden and better equipment-life management. The model should be tested and alerts verified by staff.

D2. A bank's AI loan model approves fewer loans for applicants from certain areas. What could be wrong and how should it be fixed?

Problem: the model may show **algorithmic bias**. Historical data could reflect unequal past lending, and location may act as a proxy for sensitive characteristics. A difference in approval rates does not alone prove the cause, so the bank must investigate carefully.
Corrective steps:
1. Audit training data, labels, features and outcomes for relevant groups.
2. Check error rates and approval patterns using appropriate fairness measures.
3. Remove or constrain unjustified proxy features where appropriate and improve data representation.
4. Retrain and validate the model, including tests on held-out and real-world data.
5. Explain decisions in understandable terms and offer human review and appeal.
6. Monitor outcomes after deployment and document accountability.

D3. A farmer notices spots on tomato leaves. How can AI help?

Solution: AI-assisted crop disease detection within precision agriculture.
1. Take a clear image of the affected leaf using a phone or field camera.
2. A computer-vision model trained on labelled crop images compares the visible pattern with known disease or stress examples.
3. The app reports a possible condition with uncertainty or confidence information, where available.
4. The farmer or agronomist checks the result and considers field conditions before choosing treatment.
5. If mapping tools or drones are available, affected areas can be mapped for targeted inspection or spraying where safe and legally permitted.

Benefits: earlier investigation, targeted action and potentially less unnecessary spraying. Image-based predictions can be wrong, so expert confirmation is important.

Final reminder: Learn the definition, workflow, applications, advantages and limitations for each topic. In longer answers, draw a simple flowchart and finish with a balanced conclusion. AI predictions support engineering judgement; they do not remove the need for testing, safety checks or accountability.